



MECHENG 322: Control Systems

Assignment Tracking Sheet

Student Information

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Assignment Information

Assignment Name:	Assignment	Due:	28 May at 17:00 (NZ Time)
Department:	Mechanical Engineering		
Lab / Tutorial Day:		Time:	
Lab / Tutorial Group:		Tutor:	
Notes:			

Declaration: (please read and sign)

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<http://www.auckland.ac.nz/uoa/home/about/teaching-learning/honesty/tl-uni-regs-statutes-guidelines>

I confirm that this work represents my individual/ our team's effort and does not contain plagiarised material.

I have checked the above details and verify them to be correct for the assignment I am submitting.

I am aware of the course outline regarding the format, structure and submission of this assignment(see note 2 below)

I understand that the University of Auckland takes no responsibility for lost assignments and that I agree to provide a duplicate copy if requested

I understand that uncollected assignments will be retained in secure storage until two weeks after the start of the following semester and thereafter destroyed

I agree that I will provide or submit an electronic version of my work for computerised review if requested(see Note 3 below)

Signed:  Date: 27 May 2026

Note:

Assignments are not accessible after they have been handed in. No additions/ removals will be permitted.

Marks may be withheld for students who have not submitted their work to Turnitin.com if required in the course outline.

The University of Auckland views cheating in coursework as a serious academic offence. Accordingly it may require submitted work to be reviewed against electronic source material using computerised detection mechanisms.

Q1

a(i)

Setting $d = 0$.

$$\begin{aligned}0.01\ddot{\theta} + 0.03\dot{\theta} &= T \\0.01\mathcal{L}\{\ddot{\theta}\} + 0.03\mathcal{L}\{\dot{\theta}\} &= \mathcal{L}\{T\} \\(0.01s^2 + 0.03s)\Theta(s) &= T(s) \\\dot{\Theta}(s) &= s\Theta(s) \\\frac{\dot{\Theta}(s)}{T(s)} &= \frac{s\Theta(s)}{T(s)} = s \cdot \frac{1}{0.01s^2 + 0.03s} = \frac{1}{0.01s + 0.03}\end{aligned}$$

a(ii)

$$\begin{aligned}G_{c1} \cdot G_p(s) &= (K_p + \frac{K_I}{s})(\frac{1}{0.01s + 0.03}) \\&= \frac{K_ps + K_I}{(0.01s + 0.03)s} \\CLTF &= \frac{G_s}{1 + G_sH_s} \\\text{Denominator} \Rightarrow \frac{s(0.01s + 0.03)}{s(0.01s + 0.03)} + \frac{K_ps + K_I}{s(0.01s + 0.03)} &= \frac{s(0.01s + 0.03) + K_ps + K_I}{s(0.01s + 0.03)} \\\text{Complete CLTF} \\&\frac{K_ps + K_I}{(0.01s + 0.03)s} \times \frac{s(0.01s + 0.03)}{s(0.01s + 0.03) + K_ps + K_I} = \frac{K_ps + K_I}{s(0.01s + 0.03) + K_ps + K_I} \\\frac{\dot{\Theta}(s)}{\dot{\Theta}_{ref}(s)} &= \frac{K_ps + K_I}{s(0.01s + 0.03) + K_ps + K_I} \\&= \frac{100K_ps + 100K_I}{s(s + 3) + 100K_ps + 100K_I}\end{aligned}$$

a(iii)

$$\begin{aligned}G_{rate}(s) &= \frac{100K_ps + 100K_I}{s(s + 3) + 100K_ps + 100K_I} \\\text{Towards steady state, } s \rightarrow 0 \\G_{rate}(s = 0) &= \frac{100K_I}{100K_I} = 1 \\G'_{rate}(s) &= \frac{1}{s^2 + (3 + 100K_p)s + 100K_I}\end{aligned}$$

$$\begin{aligned}
0.05 &= e^{\frac{-\zeta \cdot \pi}{\sqrt{1-\zeta^2}}} \\
\zeta &= \sqrt{\frac{\ln(0.05)^2}{(-\pi^2 + \ln(0.05)^2)}} = 0.69 \\
T_r &= \frac{0.8(\pi - \theta)}{\omega_d}, \quad \omega_d = \omega_n \sqrt{1 - \zeta^2} \\
\omega_n &= \frac{0.8(\pi - \cos^{-1}(\zeta))}{T_r \sqrt{1 - \zeta^2}} \\
&= 25.78 \text{ rad/s}
\end{aligned}$$

As the second order ODE follows the $T(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$:

$$\begin{aligned}
3 + 100K_p &= 2\zeta\omega_n \\
100K_I &= \omega_n^2 \\
K_p &= \frac{2\zeta\omega_n - 3}{100} = \frac{2(0.69)(32.22) - 3}{100} = 0.325 \\
K_I &= \frac{\omega_n^2}{100} = \frac{(25.78)^2}{100} = 6.65
\end{aligned}$$

a(iv)

The rise time and peak time remain constant. The KI gain eliminates the error, which also reduces the 2% settling time. Although Kp aids in eliminating the overshoot, KI also has an effect on the overshoot. Thus, rise time and peak time remain unchanged from changing KI.

b(i)

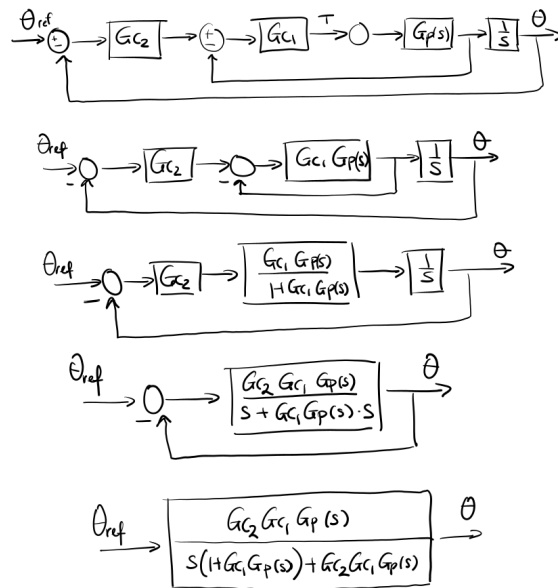


Figure 1: Theta/theta_ref working

$$\frac{G_{c2}G_{c1}G_p}{s + G_{c1}G_p \cdot s} \div \frac{s + G_{c1}G_p \cdot s + G_{c2}G_{c1}G_p}{s + G_{c1}G_p \cdot s}$$

$$CLTF = \frac{G_{c2}G_{c1}G_p}{s + G_{c1}G_p \cdot s + G_{c2}G_{c1}G_p}$$

$$\frac{\Theta(s)}{\Theta_{ref}(s)} = \frac{G_{c2}G_{c1}G_p}{s(1 + G_{c1}G_p) + G_{c2}G_{c1}G_p}$$

b(ii)

Block diagram as algebra.

$$\begin{aligned} s\Theta &= \dot{\Theta} \\ &= T \cdot G_p \\ &= (T_c + D) \cdot G_p \\ &= (((\dot{\Theta}_{ref} - \dot{\Theta})G_{c1}) + D) \cdot G_p \\ &= (((\Theta_{ref} - \Theta)G_{c2} - s\Theta)G_{c1}) + D) \cdot G_p \end{aligned}$$

As we are modelling disturbances only, set $\Theta_{ref} = 0$.

$$\begin{aligned} s\Theta &= -G_{c1}G_{c2}G_p\Theta - G_pG_{c1}\Theta + DG_p \\ \Theta[s + G_{c1}G_{c2}G_p + G_pG_{c1}] &= DG_p \\ \Theta(s) &= \frac{DG_p}{s + G_{c1}G_{c2}G_p + G_pG_{c1}} \\ \frac{\Theta(s)}{D(s)} &= \frac{G_p}{s(1 + G_pG_{c1}) + G_pG_{c1}} \end{aligned}$$

b(iii)

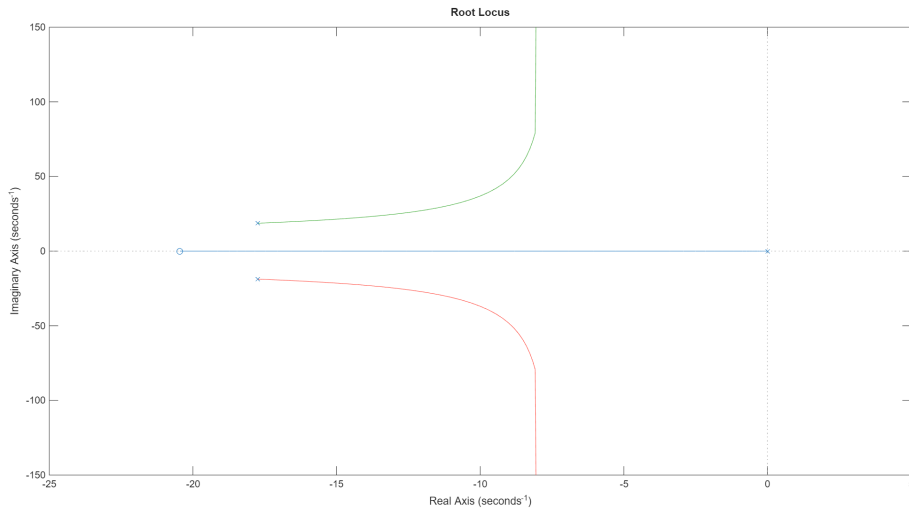


Figure 2: Root locus plot

As there is no intercept of poles with imaginary axis, the system is stable for all $K > 0$.

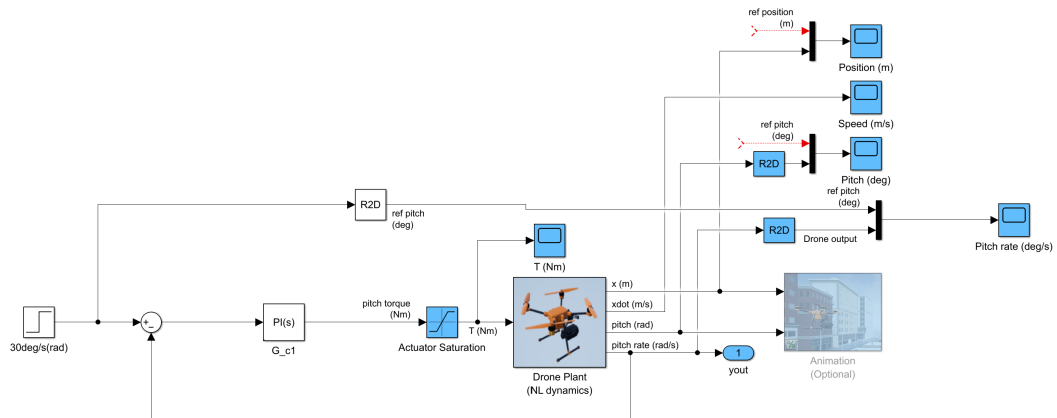
$$\mathbf{a}(\mathbf{i})$$


Figure 3: Simulink Model with pitch rate controller

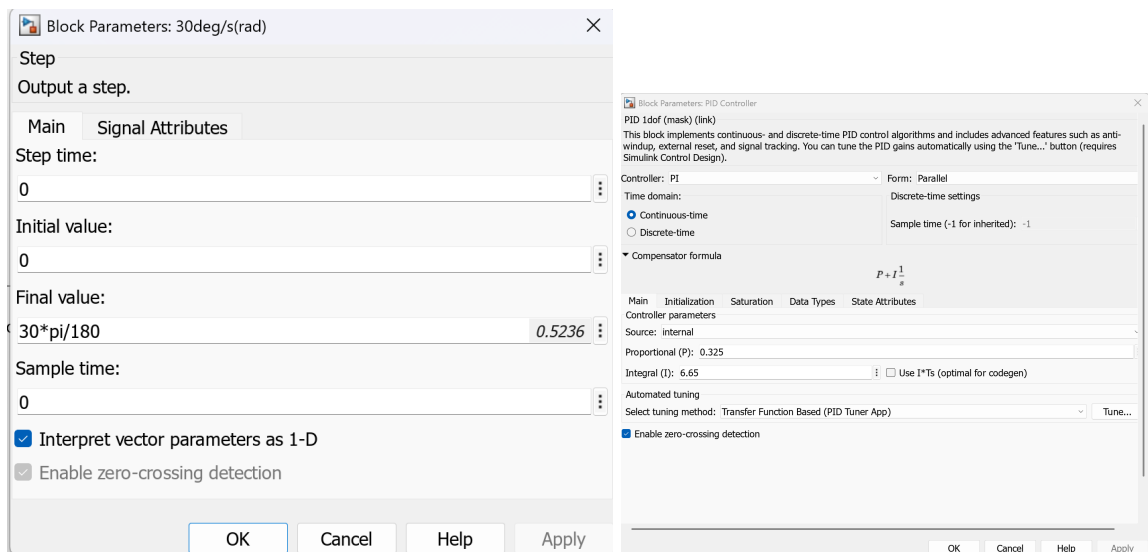


Figure 4: Parameter settings for pitch rate controller

a(ii)

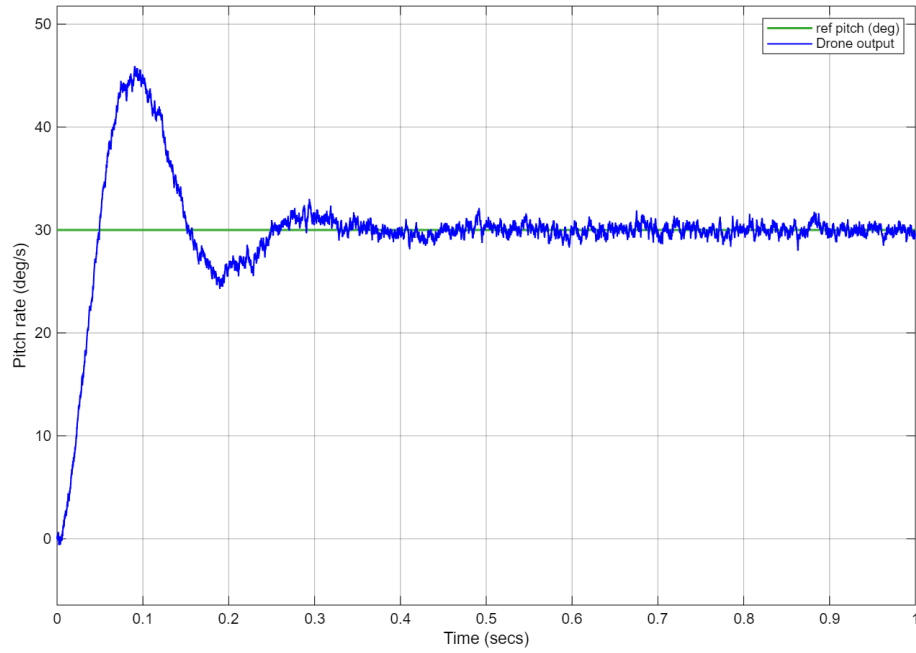


Figure 5: Pitch rate output and reference

From the provided script, the rise time (10-90%) and percentage overshoot are 0.0376 sec and 47.1%. The steady state is zero as there is an integrator term that eliminates the steady state error, and there is a pole at $s=0$.

a(iii)

The reason there is a difference in rise time compared to design requirements are due to the non linearities and disturbances from the actual drone dynamics. In Q1(a)(iii), the disturbance is ignored and is linear, which resulted in a very different output.

a(iv)

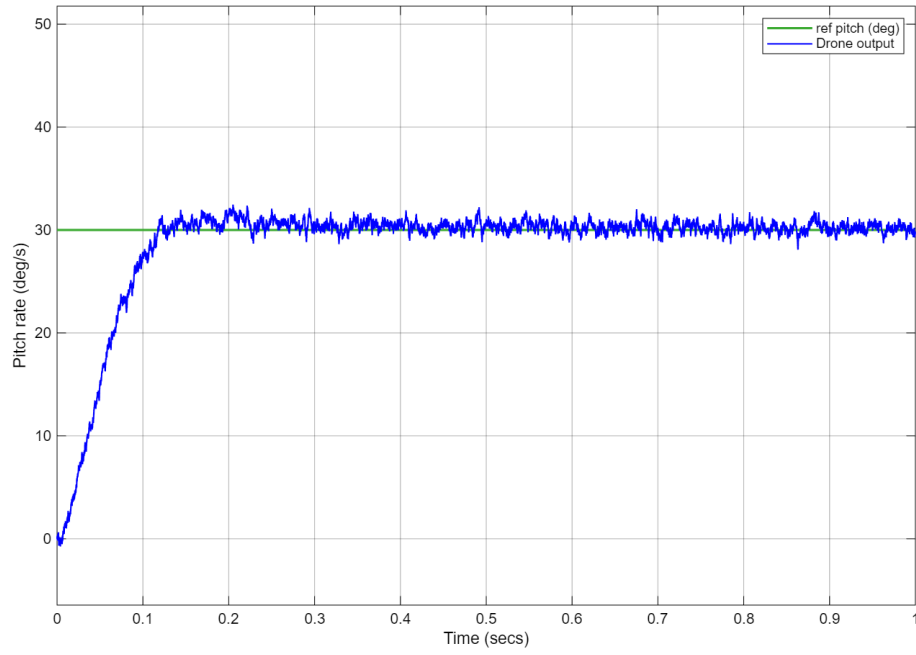


Figure 6: Tuned PI controller for pitch rate output

The first thing adjusted was the K_p value to reduce the rise time until there was minimal overshoot. K_I was then implemented to reduce the steady state error. K_p value of 0.19 and K_I value of 0.38 gave a rise time of 0.0859 seconds and percentage overshoot of 4.22%. The control law is $T(s) = (0.19 + \frac{0.38}{s}) \cdot E(s)$.

b(i)

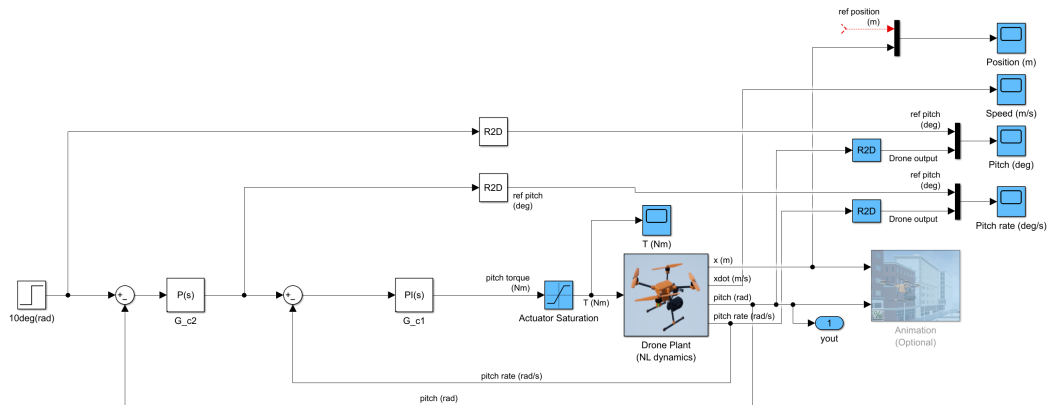


Figure 7: Simulink model with pitch controller

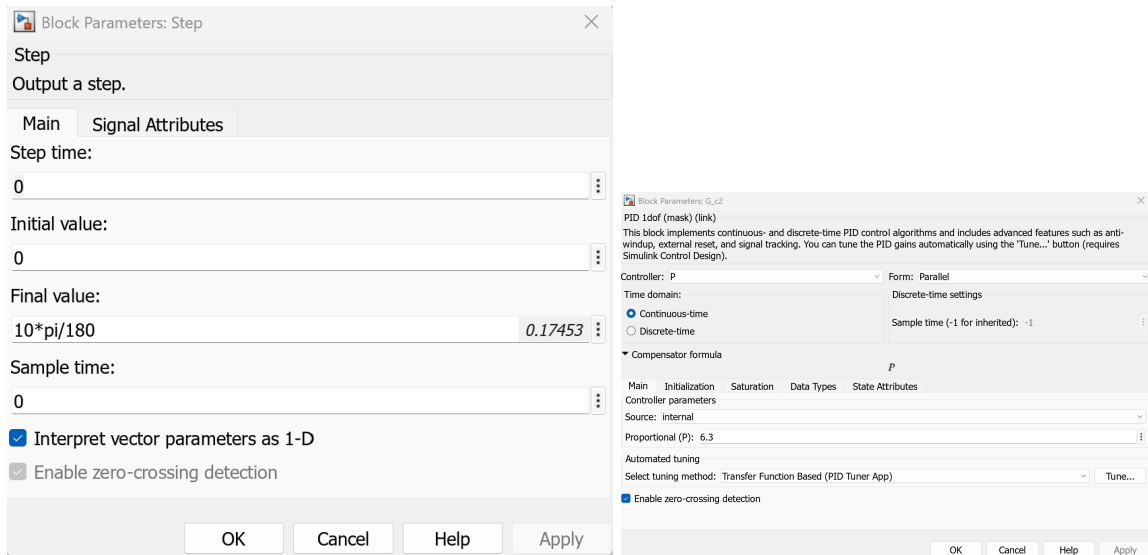


Figure 8: Parameter settings for pitch controller

b(ii)

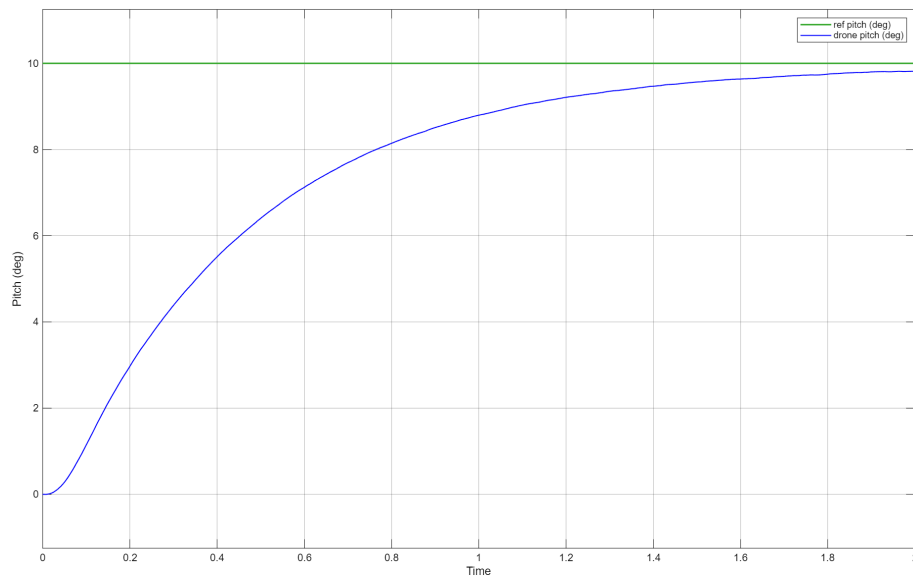


Figure 9: Output pitch angle and reference angle

The rise time and percentage overshoot are 0.97 seconds and 0%. There is a zero steady state error as there is a pole at origin ($s=0$).

b(iii)

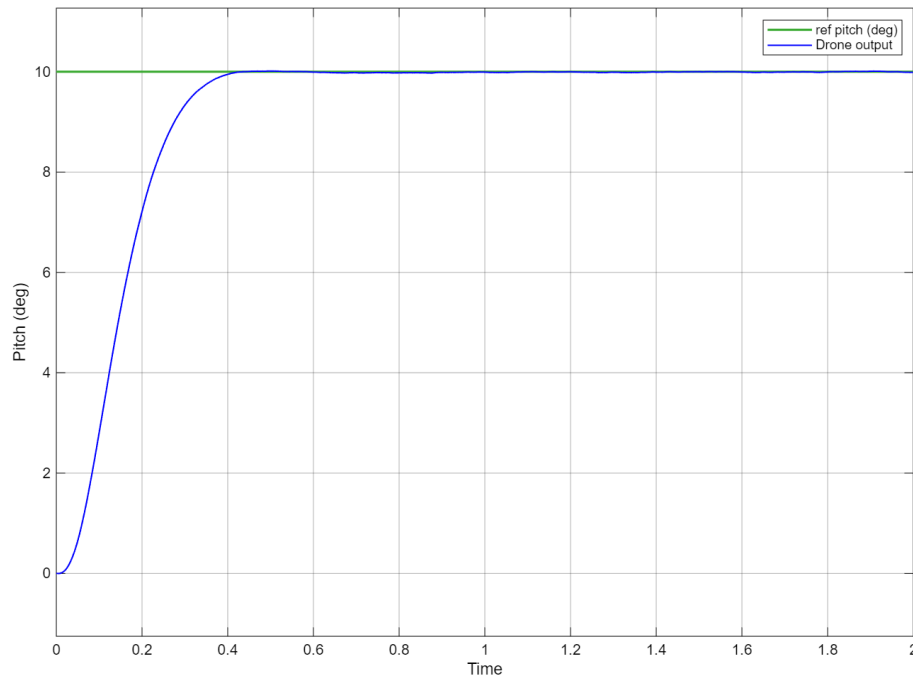


Figure 10: Tuned P controller for pitch output

The only component to adjust was the Kp value until the rise time was under 0.3 seconds. Although the rise time could get shorter with a greater Kp, it introduced oscillations, which increased percentage overshoot. To mitigate the oscillations, Kp of 6.3 was utilised, which kept rise time to a minimum of 0.217 seconds, and yielded a percentage overshoot of 0.119%.

Q3

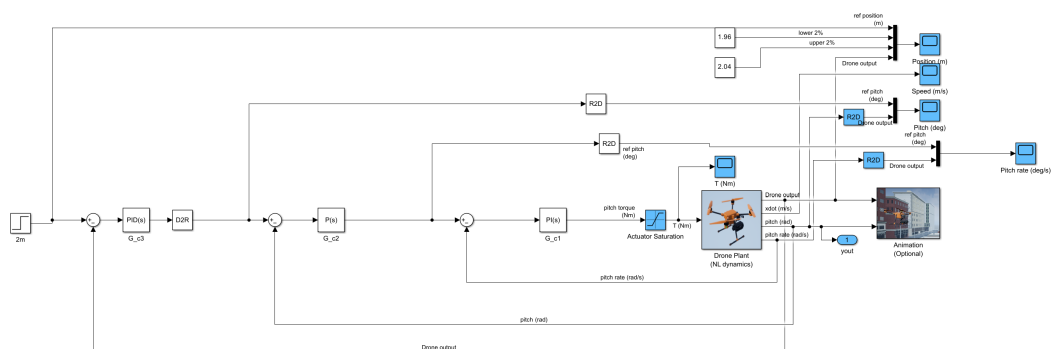


Figure 11: Simulation block

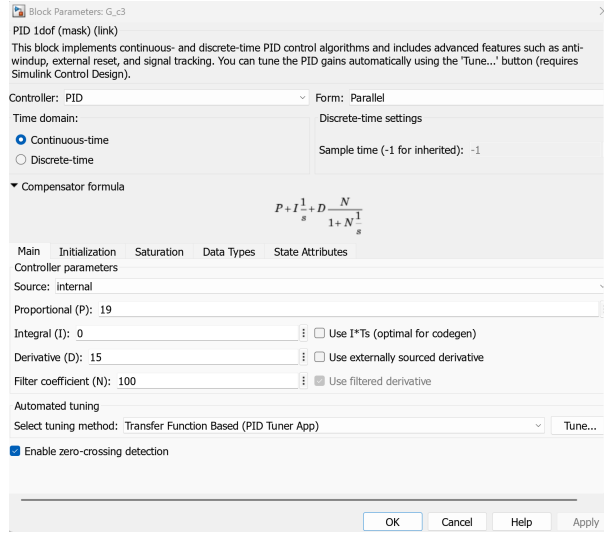


Figure 12: PID block parameters

As there was still a pole at $s=0$, the steady state error was 0, which led to $K_I=0$. Firstly, the K_p was adjusted till it could no longer be adjusted and started to be unstable. Upon reaching that value, K_d was introduced to dampen the system. As the K_d reached its maximum value, K_p had to be adjusted until the 2% settling time of under 3 seconds was met. The final values were $K_p=19$ and $K_d=15$, giving the control law as seen below.

$$\begin{aligned}\Theta_{ref}(s) &= (K_p + K_d s) \cdot (X_{ref}(s) - X(s)) \\ &= (19 + 15s) \cdot (EX(s))\end{aligned}$$

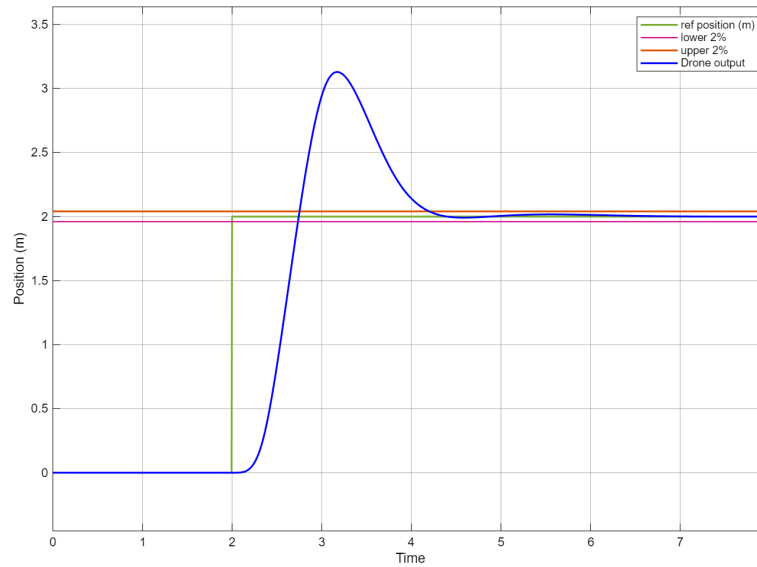


Figure 13: Position of drone with tuned PID controller

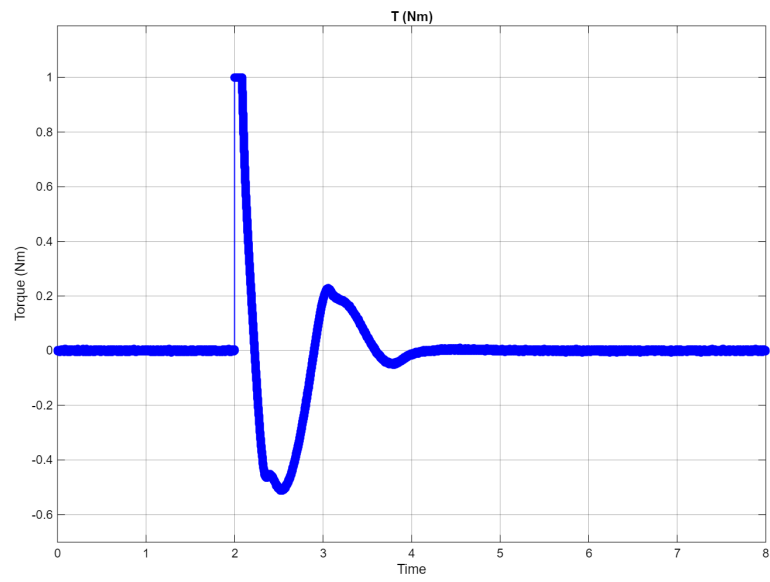


Figure 14: Pitch torque for tuned PID controller